Final Theoretical Synthesis, Protocol Specifications, and Experimental Provenance for Large-Amplitude Oscillatory Dilatational Inversion of Micellar Kinetics

Short Title: LAOD Final Synthesis & Inversion Protocols

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1. Verification of the Quadratic Amplitude Law and Invertibility Decoupling

Numerical sweeps across area oscillation strain amplitudes ranging from 0.03 to 0.30 confirm that the second-harmonic amplitude I2 obeys a strict quadratic power scaling law, with local logarithmic slopes d ln I2 / d ln epsilon between 1.84 and 2.00 [1,2]. Stepping the experimental strain from the linear limit (epsilon = 0.065) to the large-amplitude operating point (epsilon = 0.20) amplifies I2 by a factor of 9.21 (predictive target: 9.5), lifting the signal from an undetectable 0.0198 mN/m to 0.1828 mN/m, well above the 0.05 mN/m tensiometer noise floor [2].

Crucially, systematic parameter scans establish that the fundamental phase lag delta remains invariant to within 0.17 degrees across the entire strain domain (epsilon = 0.03 to 0.30). This demonstrates that the physical mechanism generating harmonic amplitude (adsorption isotherm nonlinearity) is mathematically decoupled from the mechanism generating phase lag (relaxation delay). Large deformation strain buys signal detectability at zero penalty in kinetic invertibility [1-3].

2. Experimental Provenance and Drop Mechanical Stability

A critical practical question regarding the execution of large-amplitude oscillatory dilatational rheology (LAOD) is whether a pendant drop or captive bubble remains axisymmetrically stable at epsilon = 0.20 on commercial profile analysis tensiometers [3,4].

2.1 Large-Amplitude Drop Mechanical Stability

A critical practical question regarding the execution of large-amplitude oscillatory dilatational rheology (LAOD) is whether a pendant drop or captive bubble remains axisymmetrically stable at epsilon = 0.20 on commercial profile analysis tensiometers [3,4]. Classical experimental studies by Loglio et al. and Kotsmar et al. have demonstrated that pendant

drops subjected to sinusoidal area strains up to epsilon = 0.20 to 0.30 on standard 1.0 to 1.5 mm outer-diameter capillaries maintain strict contact-line pinning without drop shedding or non-axisymmetric sloshing [3,4]. We treat epsilon = 0.20 as the recommended operational design point to be verified on the bench prior to kinetic acquisition.

2.2 Instrument Hardware Specifications and Literature Provenance

Standard commercial Sinterface PAT-1M configurations employ high-resolution CMOS/CCD cameras acquiring synchronized drop profile coordinates at 25 to 50 frames per second (PAL and USB 3.0 standards) [5]. At the 1.0 Hz operating frequency, this provides 25 to 50 frames per oscillatory cycle, far exceeding the 5 frames per cycle threshold established by Monte Carlo noise simulations [2,5]. Conversely, for fast ionic surfactant systems (CTAB and SDS), high-frequency capillary pressure tensiometry (CPT) actuated by piezoresistive micro-transducers operates across 0.1 to 500 Hz with 10 kHz data logging, providing the required high-frequency bandwidth [6].

3. Protocol Architecture for Parameter Identification

Section 7 of the manuscript defines two complementary experimental protocols for parameter extraction, explicitly acknowledging their respective strengths and statistical conditioning [2].

3.1 Protocol A: Headline Single-Concentration Inversion with Thermodynamic Closure

When the mean micellar aggregation number n_bar is known from independent light scattering or small-angle neutron scattering (SANS), the micellar capacity Cm is closed analytically via the mass-action relation Cm = ((c0 - CMC)/CMC)*n_bar [1,2]. At the operating design point, numerical evaluation yields the partial derivatives d delta / d ln Cm = +14.06 degrees per e-fold and d delta / d ln tau2 = -13.29 degrees per e-fold. Because these derivatives

possess nearly equal magnitude and opposite signs (+14.06 vs -13.29), increasing capacity and increasing relaxation time cancel in the phase lag, explaining the near-perfect collinearity (r = 0.997) at a single concentration [2].

Consequently, uncertainties in the aggregation number n_bar propagate into systematic bias on tau2 with a strict 1:1 ratio (a 10% error in n_bar produces a 10.1% systematic bias in tau2). A single oscillatory dilatational measurement at c0 = 1.5x CMC and omega = 6.28 rad/s (1.0 Hz) identifies tau2 with 0.2% to 0.3% statistical precision under 0.05 mN/m noise, subject to 1:1 systematic sensitivity to n_bar [2]. This transparently bounded result forms the primary recommendation of the study.

3.2 Protocol B: Exploratory Multi-Concentration Series Inversion

For uncharacterized surfactant systems where n_bar has not been measured, joint identification of (tau2, Cm) requires multi-concentration series regression. We transparently report that the mathematical correlation between tau2 and Cm ranges from 0.1 to 0.6 and does not decrease monotonically with the addition of arbitrary concentration points [2]. Protocol B requires a systematic 4- to 5-point sweep across 1.2x to 2.5x CMC with joint nonlinear least-squares regression, rather than a naive two-point fit.

4. Quantitative Protocol and Instrument Synthesis

Table 1 synthesizes the operational parameters, hardware requirements, and inversion protocols across the operational surfactant landscape [1,2,5,6].

Table 1. Operational parameters, instrumentation bands, and inversion protocols for micellar relaxation kinetics [2,5,6].

System Class	Model Surfactant	tau2 Scale	Operable Frequency	Primary Protocol	Ref.
Nonionic (Slow)	C12E8	1.0 to 4.0 s	0.04 to 1.19 Hz (PAT-1M)	Protocol A (0.2% stat, 1:1	[2,5]

				n_bar)		
Cationic (Medium)	CTAB	65 ms	>= 16 Hz (CPT band)	Protocol (CPT)	B	[2,6]
Anionic (Fast)	SDS	38 ms	>= 20 Hz (CPT band)	Protocol (CPT)	B	[2,6]

5. References

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